

5.2 Homework p 224 5, 13, 23, 29, 32, 37, 37

5. $\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ $\therefore \sqrt{2^2 + 2^2 + 1} = \sqrt{9} = 3$ Normalize
 $\rightarrow \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ Orthogonalize

$\text{proj}_{u_1}(v_2) = (v_2 \cdot u_1) u_1 = 3 \left(\frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$

$v_2 - \text{proj}_{u_1}(v_2) = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix} \rightarrow \sqrt{(-1)^2 + (-1)^2 + 4^2} = \sqrt{18} \Rightarrow \frac{1}{\sqrt{18}} \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix}$

$u_2 \rightarrow \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{18}} \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix}$

13. $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ $\rightarrow \sqrt{1^2 + 1^2 + 1^2 + 1^2} = \sqrt{4} = 2$ $u_1 = v_1 \rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$
 $\vec{e}_1 = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

$u_2 = \vec{v}_2 - \text{proj}_{u_1}(v_2)$

$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} \rightarrow \begin{bmatrix} 1/2 \\ -1/2 \\ -1/2 \\ 1/2 \end{bmatrix}$

$\hookrightarrow c_2 = \frac{\vec{u}_2}{|\vec{u}_2|} = \begin{bmatrix} 1/2 \\ -1/2 \\ -1/2 \\ 1/2 \end{bmatrix}$ $e_2 \rightarrow \begin{bmatrix} 1/2 \\ -1/2 \\ -1/2 \\ 1/2 \end{bmatrix}$

$\vec{u}_3 = \vec{v}_3 - \text{proj}_{u_1}(\vec{v}_3) - \text{proj}_{u_2}(\vec{v}_3)$

$\vec{u}_3 = \vec{v}_3 - \frac{\vec{v}_3 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1} \vec{u}_1 - \frac{\vec{v}_3 \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2} \vec{u}_2$

$$\vec{u}_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} - \frac{\begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\|^2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{\begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix}}{\left\| \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} \right\|^2} \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

$$\vec{u}_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} - \frac{0+2+1-1}{\sqrt{1^2+1^2+1^2+1^2}} \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{0+1-1/2-1/2}{2} \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} - \frac{2}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - 0 \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

$$\vec{u}_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} - \frac{\begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\|^2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{\begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix}}{\left\| \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} \right\|^2} \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{0}{1} \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 3/2 \\ 1/2 \\ -3/2 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -1/4 \\ 3/4 \\ 1/4 \\ -3/4 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -1/4 \\ 3/4 \\ 1/4 \\ -3/4 \end{bmatrix} \quad \text{I give up.}$$

23. QR

29.

$$\vec{v}_1 = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \quad \vec{v}_2 = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

$$\vec{u}_1 = \vec{v}_1$$

$$\vec{u}_1 = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$$

$$\vec{e}_1 = \frac{\vec{u}_1}{\|\vec{u}_1\|} = \frac{\begin{pmatrix} -3 \\ 4 \end{pmatrix}}{\sqrt{(-3)^2 + 4^2}} = \frac{1}{5} \begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix}$$

$$\vec{u}_2 = \vec{v}_2 - \text{proj}_{\vec{u}_1}(\vec{v}_2)$$

$$\vec{u}_2 = \begin{pmatrix} 1 \\ 7 \end{pmatrix} - \frac{\begin{pmatrix} 1 \\ 7 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 4 \end{pmatrix}}{\left\| \begin{pmatrix} -3 \\ 4 \end{pmatrix} \right\|^2} \begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix} - \frac{25}{25} \begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$\vec{e}_2 = \frac{\vec{u}_2}{\|\vec{u}_2\|} = \frac{\begin{pmatrix} 4 \\ 3 \end{pmatrix}}{\sqrt{4^2 + 3^2}} = \frac{\begin{pmatrix} 4 \\ 3 \end{pmatrix}}{5} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}$$

32. $x + y + z = 0$

$$z = -x - y$$

$$z = 0 \quad x = 1 \quad y = -1 \rightarrow (1, -1, 0) \quad v_1$$

$$0 = -1 + 1$$

$$z = -x - y$$

$$z = -1 \quad x = 1 \quad y = 0$$

$$-1 = -1 \rightarrow (1, 0, -1) \quad v_2$$

$$\vec{u}_1 = \vec{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$\vec{e}_1 = \frac{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}{\sqrt{1^2 + (-1)^2 + 0^2}} = \frac{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$\vec{u}_2 = \vec{v}_2 - \text{proj}_{\vec{u}_1}(\vec{v}_2)$$

$$= \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} - \frac{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}{\left\| \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\|^2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \\ -1 \end{pmatrix}$$

$$\vec{e}_2 = \frac{\begin{pmatrix} 1/2 \\ 1/2 \\ -1 \end{pmatrix}}{\sqrt{(1/2)^2 + (1/2)^2 + (-1)^2}} = \frac{\begin{pmatrix} 1/2 \\ 1/2 \\ -1 \end{pmatrix}}{\sqrt{3/2}} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$$

$$\frac{1/4 + 1/4 + 1}{\sqrt{3/2}} = \frac{3/2}{\sqrt{3/2}} = \sqrt{3/2}$$

$$33. A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{matrix} 1 \\ 2 \end{matrix}$$

$$a+b+c+d=0 \rightarrow a=-b-c-d$$

$$a+2b+3c+4d=0$$

$$-b-c-d+2b+3c+4d=0$$

$$b+2c+3d=0$$

$$b=-2c-3d$$

$$a=-2c-3d-c-d$$

$$a=-3c-4d$$

$$a=c+2d$$

$$\underline{c=s} \quad \underline{d=t}$$

$$\underline{a=s+2t} \quad \underline{b=-2s-3t}$$

$$x = [a, b, c, d] = [s+2t, -2s-3t, s, t]$$

$$x \in \text{span}(F) = (s+2t, -2s-3t, s, t)$$

$$= (s, -2s, s, 0) + (2t, -3t, 0, t)$$

$$= s(1, -2, 1, 0) + t(2, -3, 0, 1)$$

$$(2(1, -2, 1, 0) + (2, -3, 0, 1))$$

$$w_1 = v_1$$

$$|w_1|^2 = 1^2 + (-2)^2 + 1^2 + 0^2 = 1+4+1=6 \Rightarrow \sqrt{6}$$

$$e_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}$$

$$w_2 = v_2 - \frac{v_2 \cdot w_1}{w_1 \cdot w_1} w_1$$

$$v_2 \cdot w_1 = (2, -3, 0, 1) \cdot (1, -2, 1, 0) = 2+6=8$$

$$w_1 \cdot w_1 = 6$$

$$= v_2 - \frac{8}{6} v_1 = (2, -3, 0, 1) - \frac{4}{3} (1, -2, 1, 0)$$

$$w_2 = (2, -3, 0, 1) - (\frac{4}{3}, -\frac{8}{3}, \frac{4}{3}, 0) = (2 - \frac{4}{3}, -3 - (-\frac{8}{3}), 0 - \frac{4}{3}, 1 - 0)$$

$$2 - \frac{4}{3} = \frac{6}{3} - \frac{4}{3} = \frac{2}{3}, \quad -3 - (-\frac{8}{3}) = -3 + \frac{8}{3} = -\frac{9}{3} + \frac{8}{3} = -\frac{1}{3}$$

$$0 - \frac{4}{3} = -\frac{4}{3}, \quad 1 - 0 = 1$$

$$w_2 = (\frac{2}{3}, -\frac{1}{3}, -\frac{4}{3}, 1)$$

$$|w_2| = \frac{2^2}{3} + \frac{1^2}{3} + \frac{4^2}{3} + 1^2 = \sqrt{\frac{10}{3}} \Rightarrow \frac{1}{\sqrt{\frac{10}{3}}} (\frac{2}{3}, -\frac{1}{3}, -\frac{4}{3}, 1)$$

$$v_2 \Rightarrow \frac{1}{\sqrt{6}} (1, -2, 1, 0), \frac{1}{\sqrt{\frac{10}{3}}} (2, -1, -4, 3)$$